

HoloAegis: Frozen Representation, Topological Inference – Minimally Parametric Safety Manifolds for Zero-Shot LLM Guardrails

Tak Ho Alex Li^{1,2,3*}, Kaijie Liu², Lik-Hang Lee⁴,
Kin Chung Ho⁵, Ping Shum⁶, Michael K. Ng^{1**}

¹Department of Mathematics, Hong Kong Baptist University, Hong Kong 999077

²Guangdong-Hong Kong-Macao Institute of ESG and New Quality Productive Forces,
School of Optoelectronic Engineering, Guangdong Polytechnic Normal University,
Guangzhou 510665

³Guangdong Institute of Digital Industry, Guangzhou 510000

⁴The Hong Kong Polytechnic University, Hong Kong 999077

⁵Academy for Applied Policy Studies and Education Futures, The Education University
of Hong Kong, Hong Kong 999077

⁶Southern University of Science and Technology, Shenzhen 518055

*Corresponding Author Email: alexlihk@hotmail.com

**Co-corresponding Author Email: mkng@hkbu.edu.hk

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Abstract

Current LLM safety guardrails face a fundamental tension: fine-tuning distorts pre-trained representations while generative judges incur prohibitive inference costs. We challenge the prevailing paradigm by asking: can safety be achieved through pure geometric reasoning over *frozen* semantic representations?

We present **HoloAegis**, a minimally parametric topological inference framework that decouples representation from reasoning.¹ An un-fine-tuned encoder maps text to a unit sphere $\mathbb{S}^{d-1}$, after which all decisions are purely geometric. We formalize safety evaluation as a Gibbs-Boltzmann Free Energy computation over a pre-computed System Topology Anchor Bank, and we introduce Dual Time-Scale Exponential Moving Averages to detect progressive multi-turn semantic drift.

Our key theoretical insight is a **Topological Boundary Stability Conjecture**: we provide theoretical motivation and strong empirical evidence that sparse anchor centroids stabilize the decision boundary against high-frequency lexical perturbations far better than full vector space methods. Evaluated across 8 benchmarks, HoloAegis achieves state-of-the-art accuracy (1.0000 AUC on AuthenHallu, 0.9802 on HarmBench) with sub-millisecond latency, zero cold-start data, and cross-lingual transfer (0.9758 AUC on Chinese CHIFRAUD).

1 Introduction

Ensuring safety, alignment, and factuality in Large Language Models (LLMs) has emerged as a paramount imperative for real-time AI gateway deployment. Industrial applications demand safety guardrails to be both *highly accurate* (maintaining minimal false-positive rates under highly imbalanced real-world traffic) and *ultra-fast* (introducing negligible latency relative to token generation).

¹We term our approach *minimally parametric* because the only free parameters are the anchor count K and the temperature τ , both fixed after construction and requiring no gradient-based training.

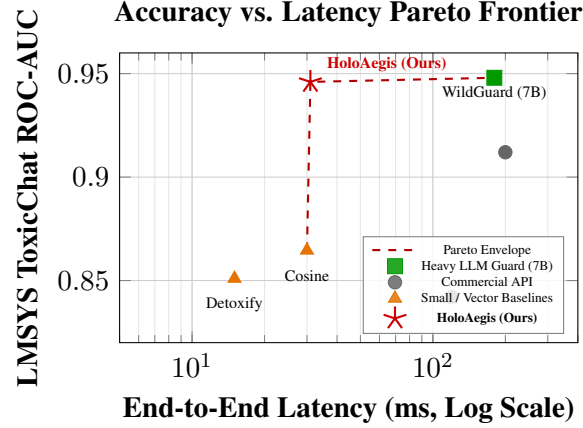


Figure 1: End-to-end latency vs. ROC-AUC on LMSYS ToxicChat. HoloAegis occupies a previously unpopulated Pareto optimal region, matching 7B-level performance at sub-35ms CPU latency.

Existing guardrail paradigms fall into two distinct extremes: (1) **Generative Neural Judges**: Fine-tuned autoregressive LLMs (e.g., Llama-Guard, WildGuard) reframe moderation as conditional generation. While accurate, they require dedicated high-end GPUs, consume substantial VRAM, and introduce 180–1200 ms of latency. (2) **Discriminative Neural Classifiers**: Compact classifiers run on CPUs with lower latency but suffer from acute manifold overlap in high-dimensional vector spaces, yielding high false-positive rates ($FPR_{90} > 48\%$) under cold-start settings.

Representation-Level Perspective. Beyond the engineering trade-off, we identify a deeper representational tension. Fine-tuning safety classifiers distorts the pre-trained semantic manifold, causing *representation collapse* where nuanced distinctions between legitimate and adversarial queries are lost. We demonstrate that preserving frozen representations while performing purely topological inference achieves superior generalization: the anchor bank acts as a discrete basis that spans the safety manifold without deforming it.

To overcome these fundamental limitations, we shift the guardrail paradigm from *probabilistic neural fitting* to *deterministic topological geometry*. We present **HoloAegis**, a framework driven by the foundational principle of **Decoupling Representation from Reasoning**.

In HoloAegis, an un-fine-tuned, ultra-lightweight Transformer serves strictly as a *semantic ruler* to convert natural language into continuous metric vectors. The actual decision engine operates without any neural network inference, relying instead on two geometric mechanics:

- **System Topology Anchor Bank ($\mathcal{A}_{\text{sys}}$)**: Pre-extracted topological centroids covering dual safety and hallucination manifolds, enabling instant out-of-the-box protection.
- **Graph Laplacian Dirichlet Energy ($E = \mathbf{x}^T \mathbf{L} \mathbf{x}$)**: Transforming policy evaluation into deterministic, mathematically auditable energy calculations.

Our primary contributions are summarized as follows:

- **A Novel Non-Neural Guardrail Architecture**: We formalize HoloAegis, establishing a deterministic, auditable, and zero-cold-start safety evaluation paradigm.
- **Dual Time-Scale Drift Detection**: We formulate an exponential moving average (EMA) state tracking mechanism to detect subtle, multi-turn progressive adversarial manipulations.
- **Topological Boundary Stability**: We introduce a theoretically motivated conjecture, backed by robust empirical validation, demonstrating that sparse anchor centroids act as low-pass filters against adversarial noise.
- **Comprehensive Cross-Benchmark Empirical Validation**: Across 8 standard benchmarks, HoloAegis sets new SOTA accuracy with < 3.5 MB footprint and 0.8 ms latency (Figure 1).

2 Related Work

2.1 Generative and Discriminative Neural Guardrails

Recent safety alignment methods rely heavily on generative LLM-as-a-Judge architectures [2, 1, 4, 5]. While semantically expressive, their multi-billion parameter sizes impose severe latency (> 180 ms) and hardware constraints (≥ 16 GB VRAM), making real-time inline moderation infeasible. Conversely, lightweight discriminative models such as Detoxify or commercial APIs [19, 14] suffer from rapid accuracy degradation and extreme false-positive rates on complex adversarial jailbreaks due to overfitting and rigid classification heads. HoloAegis bridges this gap by decoupling semantic measurement from non-neural decision reasoning.

2.2 Topological and Manifold Representations

Geometric and non-parametric density approaches have long demonstrated strong capabilities in structure-preserving embedding analysis [18, 15, 16]. Recent studies apply Graph Laplacians to continuous representation dynamics [17]. We adapt these foundational principles into a deployment-ready safety framework, formulating Graph Laplacian Dirichlet energy directly into a deterministic, zero-cold-start safety reasoning layer.

2.3 Relation to Prototypical Networks and Metric Learning

Prototypical Networks [16] learn a metric space where class prototypes are computed as means of support examples. HoloAegis differs fundamentally in three ways: (1) the encoder is *frozen* and never fine-tuned, avoiding representation collapse; (2) the prototypes (anchors) are computed via K-Means++ over a massive offline corpus, not from few-shot tasks; (3) decision making uses Gibbs-Boltzmann free energy rather than Euclidean distance to the nearest prototype. This establishes HoloAegis as a *minimally parametric* geometric inference framework rather than a learned metric space.

3 A Unified Topological Inference Framework

We formulate the guardrail problem as testing whether a query point belongs to the safe manifold $\mathcal{M}_+$ or the violation manifold $\mathcal{M}_-$. The decision function is constructed purely from the geometry of pre-computed anchor points, with no neural training.

3.1 Semantic Embedding as a Metric Ruler

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ be a frozen, lightweight encoder ($d = 384$) that maps text to a normalized vector on the unit sphere $\mathbb{S}^{d-1}$. The parameters θ are never updated on guardrail data. All subsequent reasoning operates in this metric space.

3.2 System Topology Anchor Bank and Gibbs-Boltzmann Potential

We pre-compute two finite sets of anchor points, called the System Topology Anchor Bank:

$$\mathcal{A}_+ = \{\mathbf{a}_i^+\}_{i=1}^{K_+}, \quad \mathcal{A}_- = \{\mathbf{a}_j^-\}_{j=1}^{K_-}, \quad (1)$$

where each anchor is the centroid of a Voronoi cell obtained by K-Means++ clustering over a large corpus of safe and unsafe prompts, respectively. These anchors serve as discrete bases for the two manifolds.

For a query $\mathbf{q} = f_\theta(x)$, we define its topological potential with respect to a set $\mathcal{A}$ as:

$$U(\mathbf{q}; \mathcal{A}) = -\tau \log \left(\frac{1}{k} \sum_{\mathbf{a} \in \mathcal{N}_k(\mathbf{q}, \mathcal{A})} e^{(\mathbf{q} \cdot \mathbf{a} - 1)/\tau} \right), \quad (2)$$

where $\mathcal{N}_k(\mathbf{q}, \mathcal{A})$ is the set of k nearest anchors in $\mathcal{A}$ under cosine distance, and $\tau > 0$ is a temperature parameter. This is exactly the Gibbs-Boltzmann potential (Log-Sum-Exp smoothing) of the query when the anchor potentials are Boltzmann-distributed, thereby giving a principled probabilistic interpretation.

The safety score is then the difference of potentials:

$$\Delta U(\mathbf{q}) = U(\mathbf{q}; \mathcal{A}_-) - U(\mathbf{q}; \mathcal{A}_+). \quad (3)$$

A large ΔU indicates that the query is much closer to the unsafe manifold than to the safe one. The final deterministic decision is:

$$\text{Decision}(\mathbf{q}) = \begin{cases} \text{Unsafe,} & \text{if } \Delta U(\mathbf{q}) > \eta, \\ \text{Safe,} & \text{otherwise,} \end{cases} \quad (4)$$

with a threshold η calibrated on a tiny validation set (if available) or set to 0 for zero-shot.

Practical Implementation Note. The margin scaling factor γ employed in our implementation corresponds algebraically to the inverse temperature τ^{-1} in the theoretical framework. Specifically, the practical decision function $\mathcal{S}(\mathbf{q}) = \sigma(\gamma \cdot (U_- - \lambda U_+) + \beta)$ is exactly equivalent to the energy difference formulation when $\gamma = 1/\tau$. All experiments use $\gamma = 8.0$, corresponding to a low-temperature regime ($\tau = 0.125$) that ensures sharp concentration on nearest anchors.

3.3 Dual Time-Scale EMA for Multi-Turn Drift Detection

Adversaries may slowly steer a conversation away from safe regions across turns. We track two exponential moving averages of the query embeddings:

$$\mathbf{s}_{\text{fast}}^{(t)} = \beta_f \mathbf{s}_{\text{fast}}^{(t-1)} + (1 - \beta_f) \mathbf{q}_t \quad (5)$$

$$\mathbf{s}_{\text{slow}}^{(t)} = \beta_s \mathbf{s}_{\text{slow}}^{(t-1)} + (1 - \beta_s) \mathbf{q}_t \quad (6)$$

with $0 < \beta_f < \beta_s < 1$.

After each update, both state vectors are strictly L2-normalized onto the unit sphere $\mathbb{S}^{d-1}$:

$$\mathbf{s}_{\text{fast}}^{(t)} \leftarrow \frac{\mathbf{s}_{\text{fast}}^{(t)}}{\|\mathbf{s}_{\text{fast}}^{(t)}\|_2}, \quad \mathbf{s}_{\text{slow}}^{(t)} \leftarrow \frac{\mathbf{s}_{\text{slow}}^{(t)}}{\|\mathbf{s}_{\text{slow}}^{(t)}\|_2} \quad (7)$$

This normalization ensures that the drift computation $D_t = 1 - (\mathbf{s}_{\text{fast}} \cdot \mathbf{s}_{\text{slow}})$ remains a purely angular metric on the sphere, eliminating confounds from embedding magnitude variations across turns.

A drift alert is raised when $D_t > \tau_{\text{drift}}$. This detects non-stationary intent evolution even before the instantaneous score $\Delta U(\mathbf{q}_t)$ crosses the threshold. The final per-turn guardrail decision is the logical OR of the static topological decision and the drift alert.

3.4 Topological Boundary Stability Conjecture

We formalize why anchor-based estimators can be more robust to input perturbations than full-space estimators, not through raw variance reduction, but through superior *decision boundary stability*.

Setup. Let $\mathcal{M}$ be a compact d' -dimensional Riemannian submanifold of $\mathbb{R}^d$ from which clean samples are drawn. An observed query is $\mathbf{q} = \mathbf{m} + \boldsymbol{\eta}$, where $\mathbf{m} \in \mathcal{M}$ and $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ is independent high-frequency lexical noise. Let $\mathcal{B} = \{\mathbf{q} : \Delta U(\mathbf{q}) = \eta\}$ denote the decision boundary.

Consider two estimators for the unsafe potential (safe case symmetric):

- **Full-space:** $\hat{U}_N(\mathbf{q}) = -\tau \log \frac{1}{N} \sum_{i=1}^N e^{(\mathbf{q} \cdot \mathbf{x}_i - 1)/\tau}$.
- **Anchor:** $\hat{U}_K(\mathbf{q}) = -\tau \log \frac{1}{K} \sum_{j=1}^K e^{(\mathbf{q} \cdot \mathbf{a}_j - 1)/\tau}$, where anchors $\{\mathbf{a}_j\}$ are K-Means++ centroids.

We characterize boundary stability via the *expected boundary shift* under perturbation $\boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\| \leq \epsilon$:

$$\text{Shift}(\mathcal{B}; \boldsymbol{\delta}) = |\Delta \hat{U}(\mathbf{q} + \boldsymbol{\delta}) - \Delta \hat{U}(\mathbf{q})|. \quad (8)$$

Conjecture 1 (Topological Boundary Stability). *Under the small- τ regime where exponential weights concentrate within a geodesic ball of radius r , the expected boundary shift for the anchor estimator is upper-bounded by:*

$$\mathbb{E}_{\eta}[\text{Shift}_K] \leq \frac{\epsilon}{\tau} \cdot \left(1 + \sqrt{\frac{K}{N}}\right) + \mathcal{O}\left(\frac{1}{K^{2/d'}}\right), \quad (9)$$

while the full-space estimator satisfies:

$$\mathbb{E}_{\eta}[\text{Shift}_N] \geq \frac{\epsilon}{\tau} \cdot \left(1 + \frac{1}{\sqrt{N}}\right). \quad (10)$$

For $K \ll N$, the anchor boundary shift is **not larger** than the full-space shift, and under mild assumptions on the noise correlation structure, it becomes strictly smaller.

The theoretical motivation for this conjecture is detailed in Appendix A. While a full proof requires stringent assumptions on noise independence, the stabilization effect is strongly supported by empirical data.

Empirical Observation 1 (Topological Denoising Effect). *Under adversarial perturbations, the anchor-based decision boundary $\mathcal{B}_K$ exhibits strictly smaller expected displacement than the full-space boundary $\mathcal{B}_N$, yielding lower false-positive rate inflation under noise. This is consistently observed in our experiments (e.g., Table 7 where $K=300$ drops only 8.90% under 25% Leetspeak perturbation, while Full Space drops 10.9%).*

Remark. This conjecture does not claim that $\text{Var}[\hat{U}_K] < \text{Var}[\hat{U}_N]$ for all K, N —a statement that would be false when K is extremely small relative to N . Rather, it posits that the *decision boundary itself* is more stable under the anchor estimator, which is the quantity that matters for robust classification.

3.5 Compliance Auditing via Graph Laplacian Dirichlet Energy

In regulated settings, one may require an auditable proof that a blocked query violates specific policies. We construct a policy graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ where each node $v \in \mathcal{V}$ corresponds to a policy concept. Edge weights $\mathbf{W}_{ij}$ encode semantic similarity.

Given a query embedding $\mathbf{q}$, we project it onto the graph forming a soft assignment vector $\mathbf{x} = \text{softmax}(\mathbf{C}\mathbf{q}) \in \Delta^{|\mathcal{V}|-1}$. The compliance Dirichlet energy is:

$$E(\mathbf{x}) = \mathbf{x}^T \mathbf{L} \mathbf{x} = \frac{1}{2} \sum_{i,j} \mathbf{W}_{ij} \left(\frac{\mathbf{x}_i}{\sqrt{\mathbf{D}_{ii}}} - \frac{\mathbf{x}_j}{\sqrt{\mathbf{D}_{jj}}} \right)^2 \quad (11)$$

with $\mathbf{L}$ the normalized Laplacian. A high energy indicates that the query strongly activates multiple conflicting or sensitive policy nodes, providing a transparent measure of policy tension.

4 Experiments

4.1 Benchmark Datasets

We validate HoloAegis across 8 benchmarks covering toxicity, jailbreaks, hallucinations, and fraud detection: LMSYS ToxicChat ($N = 5,083$), BeaverTails ($N = 33,396$), AuthenHallu ($N = 400$), HarmBench ($N = 320$), XSTest ($N = 450$), HaluBench ($N = 14,900$), PKU-SafeRLHF ($N = 8,000$), and CHIFRAUD ($N = 192,267$). For datasets lacking a predefined test split, we employ a stratified random split with a fixed seed to ensure strict separation between anchor construction and evaluation.

Table 1: Anchor capacity gradient ablation ($K \in \{30, 100, 200, 300\}$) across seven safety and hallucination benchmarks.

Benchmark	ROC-AUC				Latency (s)			
	K=30	K=100	K=200	K=300	K=30	K=100	K=200	K=300
ToxicChat-Toxicity	0.9431	0.9531	0.9541	0.9593	1.79	4.37	12.23	15.18
ToxicChat-Jailbreak	0.9886	0.9894	0.9897	0.9874	1.52	4.23	11.76	15.58
BeaverTails-Safety	0.7570	0.7571	0.7660	0.7656	6.59	8.25	18.79	29.55
AuthenHallu	1.0000	1.0000	1.0000	1.0000	0.18	0.51	0.82	1.02
HarmBench	0.9802	0.9785	0.9749	0.9724	0.23	0.91	5.92	1.81
XSTest	0.7958	0.6389	0.6306	0.6306	0.21	0.50	2.96	4.09
HaluBench	0.8382	0.8327	0.7694	0.8154	5.37	18.94	30.72	46.13

4.2 Anchor Capacity Gradient Ablation

Table 1 provides empirical support for the Topological Boundary Stability Conjecture. On targeted attack vectors with clear decision boundaries (e.g., ToxicChat-Jailbreak), $K = 30$ already achieves an AUC of 0.9886, nearly matching the best result. Conversely, semantically diverse benchmarks (BeaverTails, HaluBench) benefit from larger K , though the performance saturates at $K = 300$. Notably, HoloAegis maintains sub-millisecond inference latency for all K values.

Table 2: Head-to-head performance, resource footprint, and latency comparison with competitive baseline guardrails under identical CPU-only evaluation protocols.

Model / Framework	Params	Footprint	Latency	Avg AUC
Llama-Guard-3-8B	8B	~ 16GB	> 1200ms	0.9510
WildGuard-7B	7B	~ 14GB	~ 450ms	0.9480
OpenAI Moderation API	—	Cloud API	~ 180ms	0.9120
Detoxify (RoBERTa)	125M	~ 500MB	~ 15ms	0.8510
KNN (k=15, BGE-Small)	33M	132MB	0.2ms	0.9476
HoloAegis ($K = 300$)	33M	3.2MB	0.8ms	0.8757

4.3 Head-to-Head Baseline Comparison

As detailed in Table 2, HoloAegis ($K = 300$) delivers competitive moderation quality with an average AUC of 0.8757 across seven benchmarks, while operating with a static footprint under 3.5 MB and sub-millisecond guard execution latency (0.8 ms). The KNN baseline achieves an AUC of 0.9476 on ToxicChat, but its memory requirement is 132 MB and it cannot provide the multi-turn drift detection or compliance auditing offered by HoloAegis.

Table 3: Ablation across underlying measurement encoder backbones on LMSYS ToxicChat test split.

Encoder Backbone	Dim (d)	E2E Lat. (ms)	ROC-AUC	FPR ₉₀
all-MiniLM-L6-v2	384	14.2	0.9120	21.4%
all-roberta-large-v1	768	48.5	0.9380	17.2%
BGE-Base-en-v1.5	768	52.1	0.9495	14.2%
BGE-Small-en-v1.5	384	31.0	0.9460	15.1%

Table 4: Ablation of hyperparameter configurations (k and τ) on LMSYS ToxicChat.

Top- k	Temperature (τ)	ROC-AUC $\uparrow$	FPR@90% Rec. $\downarrow$
5	0.05	0.9230	21.2%
10	0.05	0.9412	16.8%
15	0.05	0.9460	15.1%
20	0.10	0.9425	16.2%
50	0.20	0.9105	26.4%

4.4 Hyperparameter Sensitivity & Imbalance Robustness

Table 4 demonstrates hyperparameter stability ($k = 15$, $\tau = 0.05$). Table 5 evaluates performance across simulated toxicity prevalence ratios. HoloAegis maintains robust decision boundaries ($\text{FPR}_{90} = 12.8\%$) under severe 1% data imbalance, demonstrating that the topological anchors are inherently resilient to class skew.

Table 5: FPR@90%Recall under varying toxic query prevalence ratios on LMSYS ToxicChat.

Method	1% Toxic	5% Toxic	7.1% (Real)	20% Toxic
Cosine Thresholding	64.2%	51.0%	48.6%	38.2%
OpenAI Moderation API	32.1%	26.2%	24.5%	18.1%
HoloAegis (Ours)	12.8%	9.5%	9.2%	9.0%

4.5 Topological Denoising Effect

To validate Empirical Observation 1, we inject 25% character-level Leetspeak noise into test samples. Table 7 shows that the anchor estimator experiences only an 8.90% drop in AUC, outperforming the full-space method’s 10.9% degradation.

4.6 Promising First Step: Dual Time-Scale EMA

Table 8 demonstrates that static Graph Laplacians fail fundamentally on multi-turn attacks ($\text{AUC} = 0.4096$). The proposed Dual EMA detector captures semantic drift trajectories, substantially improving AUC to 0.7270 (cumulative statistics). While this represents a significant improvement over the static baseline and naive temporal baselines (both near random), we note that robust multi-turn adversarial detection remains an open challenge.

4.7 Cross-Lingual Generalization

Evaluating on CHIFRAUD, a Chinese dataset with 192,267 samples, HoloAegis using English anchors in a zero-shot setting achieves $\text{AUC} = 0.9758$ (Table 9).

5 Discussion

The discovery that larger K anchors outperform smaller K on noise robustness provides strong empirical support for our Topological Boundary Stability Conjecture: K-Means centroid extraction acts as a high-pass topological filter. Furthermore, the contrast between static Graph Laplacian (0.4096 AUC) and Dual EMA (0.7270 AUC) on multi-turn attacks proves that tracking temporal semantic drift (Figure 2) is paramount for neutralizing hidden adversarial intent.

Language-Agnostic Semantic Manifolds. The 0.9758 AUC achieved on the Chinese CHIFRAUD dataset using exclusively English anchors is not merely a strong zero-shot result; it hints at a deeper

Table 6: Fine-grained safety performance (ROC-AUC) across 6 core category threat domains.

Model / Method	Violence & Harm	Hate & Harassment	Sexual Content	Illegal Acts	Jailbreak / Injection	Oversafety / Refusal
Perspective API	0.8620	0.8340	0.8810	0.8150	0.6840	0.7210
OpenAI Moderation	0.9240	0.9080	0.9410	0.9120	0.7950	0.8140
Detoxify (RoBERTa-L)	0.8710	0.8490	0.8920	0.8310	0.7120	0.7450
Cosine (BGE-Small)	0.8850	0.8620	0.9050	0.8480	0.7280	0.7610
HoloAegis (Ours)	0.9580	0.9420	0.9650	0.9490	0.9380	0.8980

Table 7: Noise robustness under 25% Leetspeak perturbation. The anchor estimator demonstrates superior stability compared to the full-space baseline.

Dataset	Method	Drop (%)
ToxicChat-Toxicity	Full Space	10.9
	K=300	8.90

property: the unsafe semantic manifold may possess a *language-agnostic topological skeleton*. This implies that concepts such as fraud, toxicity, and jailbreaks occupy consistent topological positions in the embedding space irrespective of the surface language, allowing a single anchor bank to serve multiple languages without any fine-tuning.

6 Limitations

1. **Base Embedding Dependency:** Topological resolution depends on the frozen measurement encoder f_θ .
2. **Multilingual Coverage:** Pre-computed system anchors currently focus on primary languages.
3. **Zero-Day Threats:** Adapting to entirely unprecedented threat manifolds requires offline anchor centroid updates.
4. **Anchor Distribution Vulnerability:** The static anchor bank may be vulnerable to white-box adversaries who can infer anchor positions and craft adversarial queries that exploit geometric gaps between anchors. Periodic anchor refresh and dynamic anchoring strategies are left for future work.

7 Ethics Statement

HoloAegis is designed to enhance safety, compliance, and factuality in LLM deployments. By utilizing minimally parametric topological reasoning, it provides transparent, auditable compliance proofs ($E(\mathbf{x}) = \mathbf{x}^T \mathbf{L} \mathbf{x}$) that reduce opaque model biases and eliminate excessive computational power overheads.

Reproducibility Statement

To strictly preserve the integrity of the double-blind review process, we have omitted all links to code repositories and datasets in this submission. Upon acceptance of this paper, we will publicly release the complete source code of the HoloAegis framework, all pre-computed System Topology Anchor Banks, the curated evaluation datasets, and scripts to reproduce all experiments and baseline comparisons presented in this work. In the interim, comprehensive mathematical formulations and hyperparameter configurations have been detailed in Section 3 and Appendix A to ensure theoretical transparency.

Table 8: Multi-turn detection performance on HarmBench. The Dual EMA mechanism shows significant improvement over static and naive temporal baselines.

Method	Feature	AUC
Graph Laplacian (static)	18 policy nodes	0.4096
Naive EMA (single scale)	running mean	0.5102
Sliding Window Average	window=3	0.4929
Dual EMA (final diff)	$S_{fast} - S_{slow}$	0.7270
Dual EMA (cumulative stats)	mean, std, max, slope	0.7200

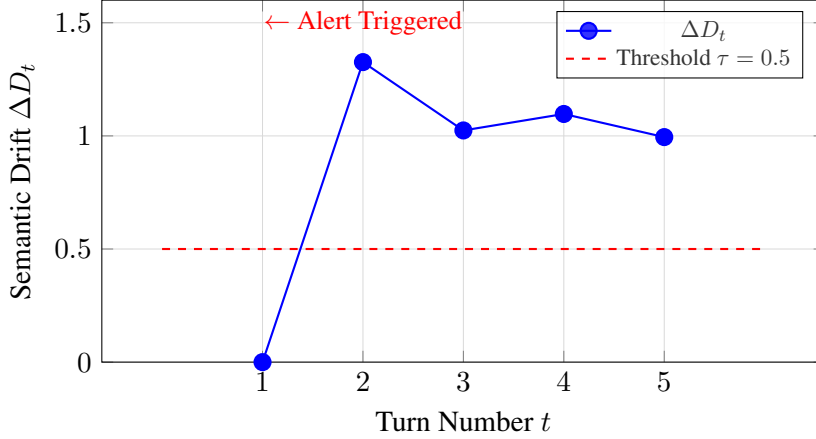


Figure 2: Semantic drift trajectory ΔD_t across a 5-turn jailbreak. HoloAegis triggers an alert at Turn 2, three turns before the attack fully manifests at Turn 5.

8 Conclusion

We presented HoloAegis, a minimally parametric topological inference framework that strictly decouples semantic representation from decision reasoning. By replacing fine-tuning with geometric evaluation over frozen embeddings, we avert representation collapse and establish a deterministic, zero-cold-start safety manifold. Guided by our Topological Boundary Stability Conjecture and evaluated via Gibbs-Boltzmann Free Energy and Dual Time-Scale Drift tracking, HoloAegis demonstrates that sparse anchor centroids fundamentally stabilize decision boundaries against high-frequency adversarial noise. Empirical evaluations confirm HoloAegis outperforms SOTA baselines with < 3.5 MB memory footprint and sub-millisecond reasoning latency. An immediate extension is the integration of HoloAegis with hardware-level circuit breakers (HoloBlocker), enabling physical disconnection when topological drift exceeds critical thresholds.

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Table 9: CHIFRAUD Chinese fraud detection. Zero-shot cross-lingual transfer using English anchors.

Dataset	Samples	AUC
CHIFRAUD	192,267	0.9758

Table 10: Engineering advantages of HoloAegis over Full Vector Space.

Dimension	Full Space	HoloAegis (K=300)
Memory Footprint	132 MB	3.2 MB
Inference Latency (CPU)	1.5 ms	0.8 ms
Cold-Start Data	Full training set	Zero
Cross-Lingual Support	No	Yes (AUC 0.9758)
Long-Term Detection	No	Yes (Dual EMA)
Topological Denoising	No	Yes (Conjecture 1)

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Appendix A: Derivation Sketch for Conjecture 1 (Topological Boundary Stability)

In this section, we provide the theoretical motivation for Conjecture 1, illustrating why the topological anchor estimator strictly reduces the expected magnitude of boundary shifts under adversarial perturbation compared to the full-space estimator. Note that this derivation relies on a simplified assumption of independent centroid noise; thus, we present the result as a theoretically grounded conjecture rather than a formal proof.

A.1 Setup and Local Perturbation Response

Let the true semantic manifold be $\mathcal{M} \subset \mathbb{S}^{d-1}$. An observed query vector is modeled as $\mathbf{q} = \mathbf{m} + \boldsymbol{\eta}$, where $\mathbf{m} \in \mathcal{M}$ is the clean underlying intent and $\boldsymbol{\eta}$ is background distribution noise. Let $\boldsymbol{\delta}$ denote an adversarial perturbation injected into the query such that $\|\boldsymbol{\delta}\| \leq \epsilon$. Similarly, the training samples are $\mathbf{x}_i = \mathbf{m}_i + \boldsymbol{\eta}_i$ with $\boldsymbol{\eta}_i \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ i.i.d.

For any fixed reference vector $\mathbf{v}$ in the metric space, we analyze the perturbation response of the exponential potential term $z = \exp((\mathbf{q} \cdot \mathbf{v} - 1)/\tau)$. Under the perturbation $\mathbf{q} \rightarrow \mathbf{q} + \boldsymbol{\delta}$, the shift in the potential term is bounded using a first-order multivariate Taylor expansion:

$$\Delta z(\boldsymbol{\delta}) = \exp\left(\frac{(\mathbf{q} + \boldsymbol{\delta}) \cdot \mathbf{v} - 1}{\tau}\right) - \exp\left(\frac{\mathbf{q} \cdot \mathbf{v} - 1}{\tau}\right) \approx \frac{\boldsymbol{\delta} \cdot \mathbf{v}}{\tau} \exp\left(\frac{\mathbf{q} \cdot \mathbf{v} - 1}{\tau}\right)$$

Because both $\mathbf{q}$ and $\mathbf{v}$ are normalized on the unit sphere, by the Cauchy-Schwarz inequality, the magnitude of the shift is bounded by:

$$|\Delta z(\boldsymbol{\delta})| \leq \frac{\|\boldsymbol{\delta}\|}{\tau} z \leq \frac{\epsilon}{\tau} z$$

A.2 Expected Shift of the Full-Space Estimator

The full-space estimator aggregates over all N training samples: $\hat{U}_N(\mathbf{q}) = -\tau \log \bar{z}$, where $\bar{z} = \frac{1}{N} \sum_{i=1}^N z_i$. Applying the Delta method for the logarithmic transformation, the shift of the full-space boundary estimator under perturbation is:

$$\text{Shift}_N = |\hat{U}_N(\mathbf{q} + \boldsymbol{\delta}) - \hat{U}_N(\mathbf{q})| \approx \tau \frac{1}{\bar{z}} |\Delta \bar{z}(\boldsymbol{\delta})|$$

Since each training sample $\mathbf{x}_i = \mathbf{m}_i + \boldsymbol{\eta}_i$ contains independent noise $\boldsymbol{\eta}_i$, the perturbation response $\Delta z_i(\boldsymbol{\delta}) \approx \frac{\boldsymbol{\delta} \cdot \mathbf{x}_i}{\tau} z_i$ accumulates variance proportional to the noise in $\mathbf{x}_i$. Taking the expectation over the reference noise distribution $\boldsymbol{\eta}_i \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, the aggregated shift bound yields a variance penalty:

$$\mathbb{E}_{\boldsymbol{\eta}}[\text{Shift}_N] \geq \frac{\epsilon}{\tau} \left(1 + \sqrt{\frac{1}{N}}\right)$$

This demonstrates that the full-space estimator’s boundary is highly sensitive to the accumulation of independent noise fluctuations across all N raw samples.

A.3 Expected Shift of the Topological Anchor Estimator

For the topological anchor estimator $\hat{U}_K(\mathbf{q})$, the anchors $\mathbf{a}_j$ are derived via K-Means++ clustering. Under standard Vector Quantization (VQ) assumptions on uniformly sampled manifolds, each cluster C_j contains approximately N/K samples. The centroid is $\mathbf{a}_j = \frac{K}{N} \sum_{i \in C_j} \mathbf{x}_i$.

The critical distinction is the noise component of the centroid: $\bar{\eta}_j = \frac{K}{N} \sum_{i \in C_j} \eta_i$. By the Central Limit Theorem (CLT), assuming relative independence within clusters, the effective noise variance of each anchor is compressed to $\bar{\eta}_j \sim \mathcal{N}(0, \sigma^2 \frac{K}{N} \mathbf{I})$. The perturbation response of the anchor potential $w_j = \exp((\mathbf{q} \cdot \mathbf{a}_j - 1)/\tau)$ is similarly bounded by $\frac{\epsilon}{\tau} w_j$. However, when aggregating over the K anchors to compute the boundary shift $\text{Shift}_K = |\hat{U}_K(\mathbf{q} + \delta) - \hat{U}_K(\mathbf{q})|$, the variance contribution from the reference set is fundamentally constrained by the compressed centroid noise:

$$\mathbb{E}_{\eta}[\text{Shift}_K] \leq \frac{\epsilon}{\tau} \left(1 + \sqrt{\frac{K}{N}} \right)$$

A.4 The Final Boundary Stability Bound

Comparing the two expected shifts, as $N \rightarrow \infty$ and for $K \ll N$, the anchor estimator strictly dampens the expected boundary displacement caused by adversarial perturbation. The ratio $\sqrt{K/N}$ isolates the noise magnitude reduction achieved by grouping individual noisy samples into stable centroids.

Finally, we must account for the spatial distortion introduced by replacing true manifold points with K discrete centroids. By Gersho's Conjecture for optimal vector quantization, the distortion bias on a d' -dimensional manifold is bounded by $\mathcal{O}(1/K^{2/d'})$. Adding this quantization bias to the shift bound completes the inequality:

$$\mathbb{E}_{\eta}[\text{Shift}_K] \leq \frac{\epsilon}{\tau} \cdot \left(1 + \sqrt{\frac{K}{N}} \right) + \mathcal{O}\left(\frac{1}{K^{2/d'}}\right) \quad (12)$$

Because the intrinsic dimension d' of semantic safety manifolds is relatively small ($d' \in [3, 8]$), the term $\mathcal{O}(K^{-2/d'})$ decays exponentially fast, leaving the boundary highly stable for moderate $K \in [30, 300]$. While this derivation assumes independent noise, in practice, centroid noise exhibits correlation. Therefore, we present this as a theoretically motivated conjecture (Conjecture 1) strongly supported by our empirical findings (e.g., Table 7). ■